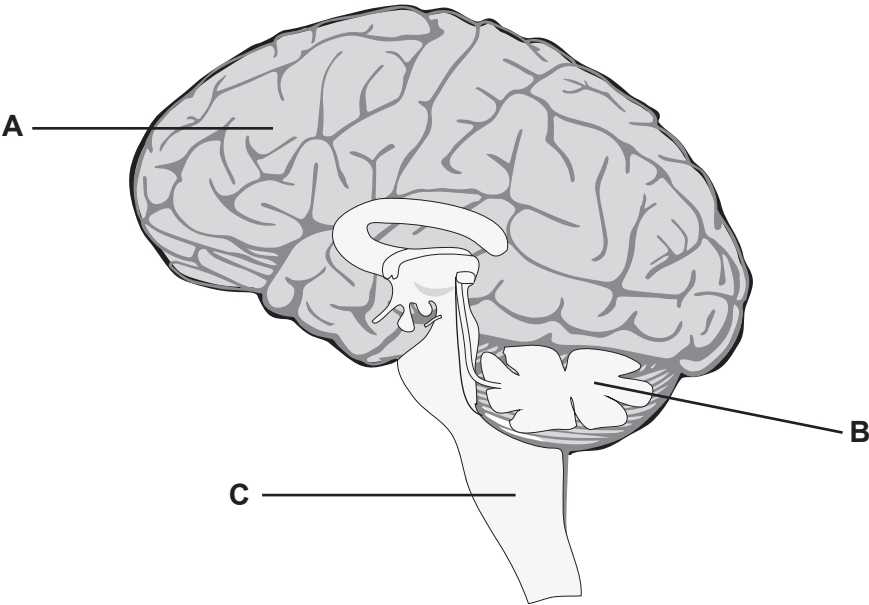


Option C: Neurobiology and Behaviour

9. The diagram below shows a section through the human brain.



(a) Identify **A**, **B** and **C** on the diagram above.

[1]

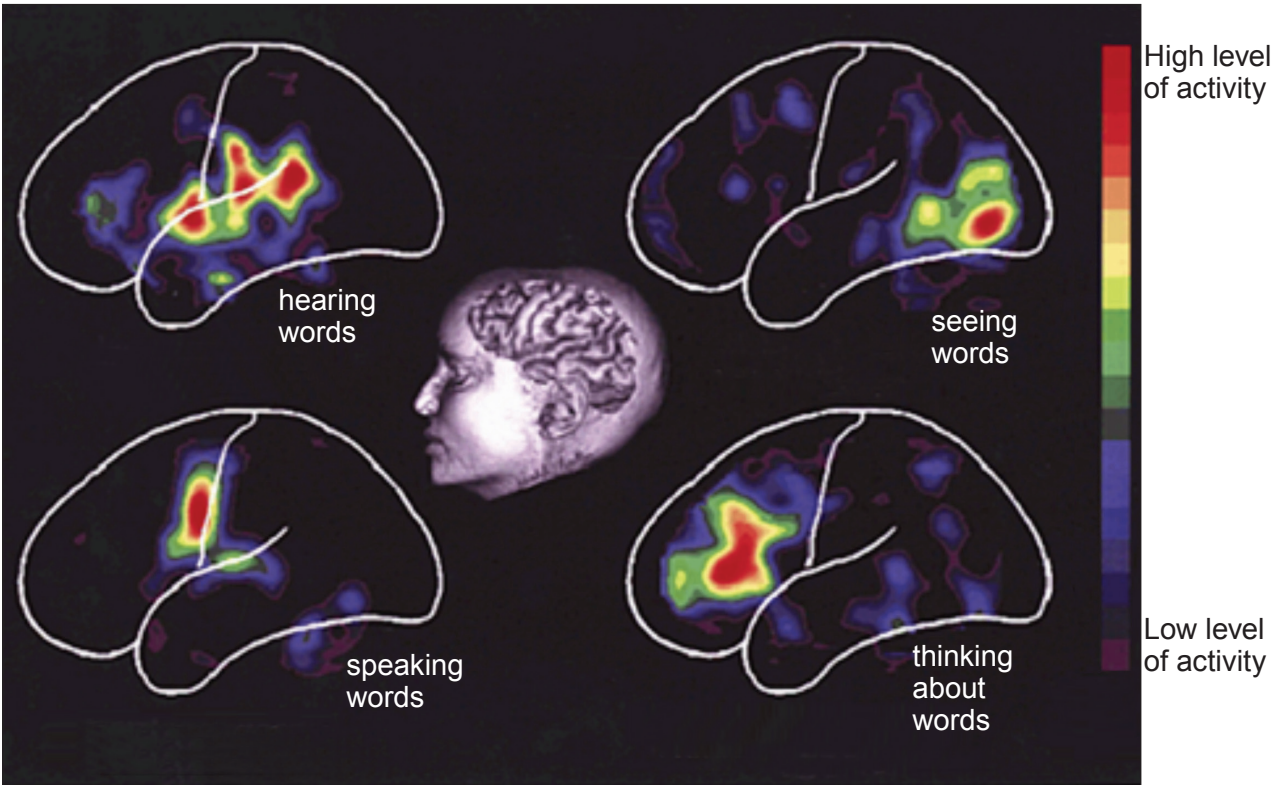
- A**
- B**
- C**



- (b) It is estimated that there are 125 000 deaf adults in the UK who use British Sign Language (BSL). The language involves movement of the hands, body, face and head. Some people are born deaf and others become deaf during their lives. There are many causes of deafness, including damage to the auditory nerve between the ear and the brain.

Positron emission tomography (PET) can be used to generate images of the brain using radioactive tracers which have a short half-life. There is a PET scanner at Cardiff University.

The image below shows the results of a PET scan when a hearing patient was given tasks associated with language.



- (i) Explain why a PET scan would be suitable for investigating activity levels within the brain whereas an MRI scan would not. [2]

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Examiner
only

- (ii) With reference to the image of the PET scan and your knowledge of brain regions involved in language, suggest and explain how the PET scan of a deaf person, actively using BSL, would compare to that of a hearing person. [5]

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- (iii) Suggest **two** medical details that would be needed when interpreting PET scans from patients with hearing problems. [2]

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- (iv) Explain why PET scans of adults with total hearing loss have revealed activation of regions of the cortex related to hearing. [3]

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- (c) The three-spined stickleback, *Gasterosteus aculeatus*, is common to lakes and rivers in Wales. The male stickleback has a red belly during the breeding season and behaves aggressively when defending its territory.

The behaviour of 12 male sticklebacks was investigated during the breeding season. Each male stickleback was exposed to model sticklebacks with red or silver bellies and the number of times each model was bitten was recorded.



Male and female sticklebacks during the breeding season, in Llyn Frongoch, Ceredigion

- (i) State **one** advantage to male sticklebacks of defending their territory. [1]

The table below shows the results of the investigation.

Model colour	Mean number of bites
red belly	119.8
silver belly	60.4

- (ii) State what is meant by a **sign stimulus** and based on this information, identify what it would be for the stickleback. Explain your answer. [3]

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Model colour	Mean number of bites	$\sum(x - \bar{x})^2$	Standard Deviation
red belly	119.8	524.7	6.9
silver belly	60.4	343.3	

- (iii) **Complete** the table above by calculating the standard deviation for the silver belly stickleback using the formula below: [2]

$$\text{Standard Deviation} = \sqrt{\frac{\sum(x - \bar{x})^2}{N - 1}}$$

where:

$\bar{x}$ = mean
 $\sum$ = sum of
N = number of samples

- (iv) The standard deviation for both models was high. Explain how this could affect your confidence in the conclusion. [1]

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END OF PAPER

